

Assignment 2 – AI-Powered Business Data Analysis with Julius AI

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Platform: <https://www.julius.ai>

Lab 1: Explore and Clone an Existing Julius AI App (EDA)

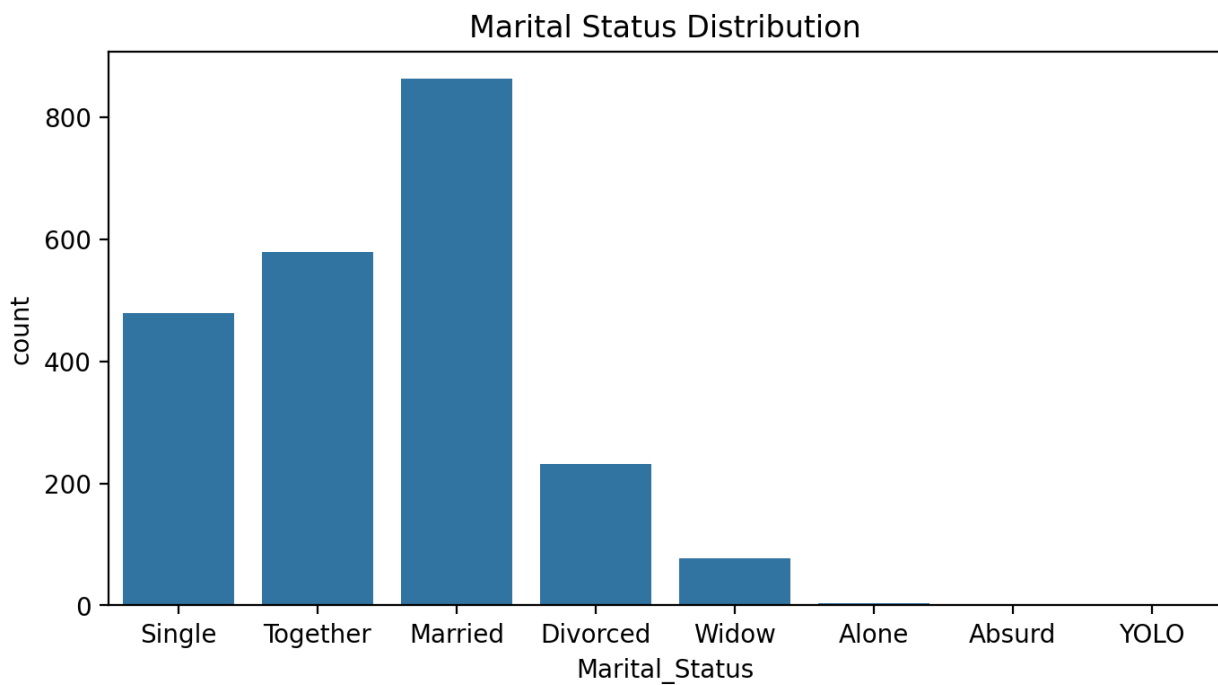
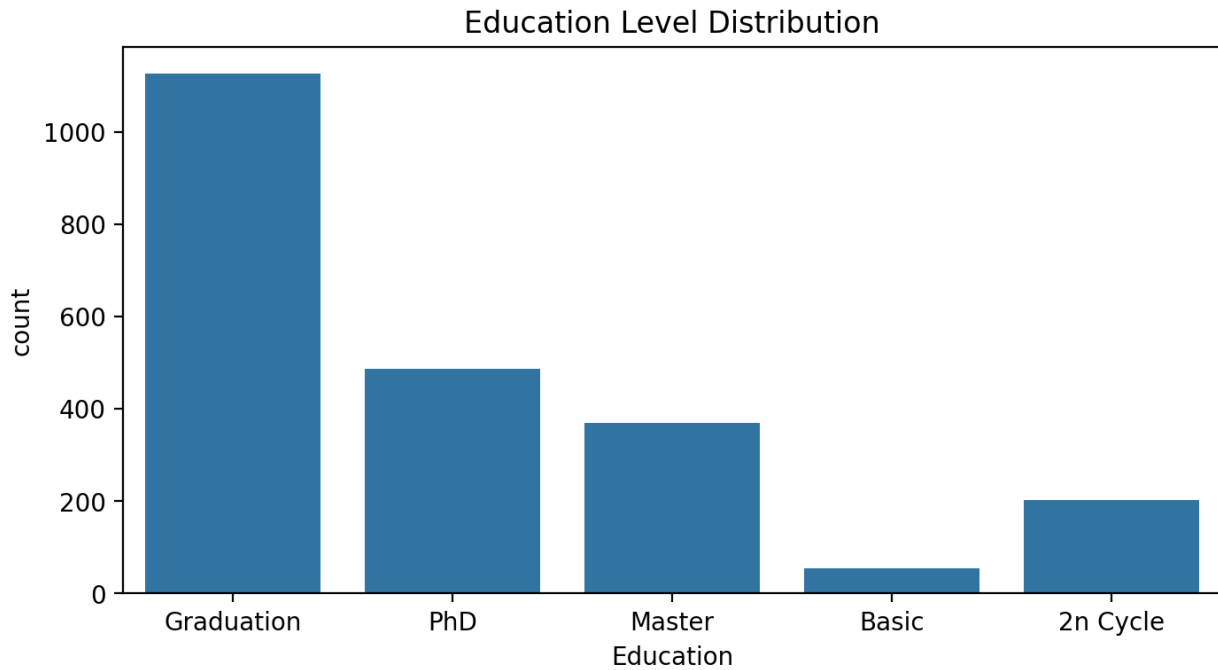
1. Find a dataset

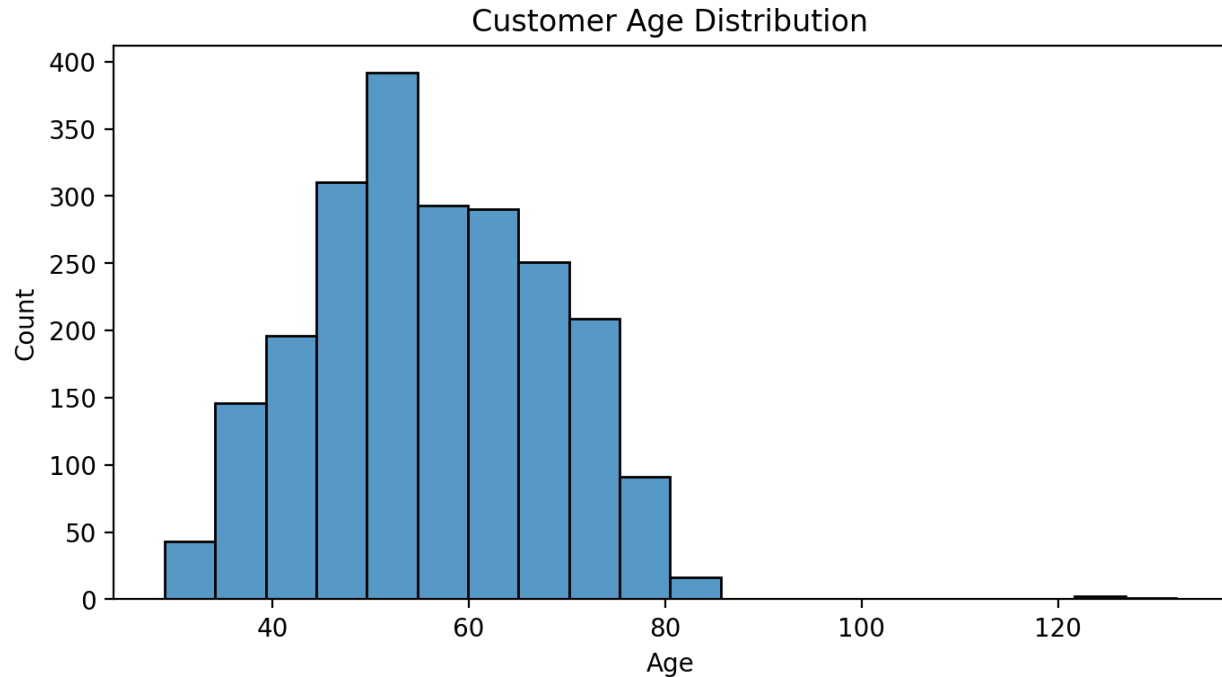
Dataset: marketing_campaign.csv

2. Analyze the Notebook

- **What is the notebook analyzing?** The dataset is analyzing customer information and their purchasing behavior. It includes demographic details such as age, education, and marital status, as well as various purchase-related metrics like spending on wines, fruits, meat, fish, sweets, and gold. Additionally, it tracks customer interactions with marketing campaigns, web visits, and responses to promotions. This data can be used to understand customer profiles, segment the customer base, and evaluate the effectiveness of marketing strategies. For example, we can analyze which demographic groups are more responsive to campaigns or which products are most popular among different segments.
- **What type of data is used (CSV, Excel, Google Sheet)?** CSV
- **What kinds of insights or visualizations does it produce?**

The visualizations show the distribution of customer demographics in the dataset. The education level distribution indicates the most common education categories among customers, while the marital status distribution reveals the proportion of singles versus those in relationships. The age distribution provides insight into the typical age range of the customer base, which can inform targeted marketing strategies.





These charts suggest that the majority of customers have a graduation or higher education level, with a significant portion being single. The age distribution peaks around the late 30s to early 50s, indicating a middle-aged customer base. This demographic profile can help tailor marketing efforts and product offerings to better suit the audience.

- **What prompts were used to drive analysis?**

What is the dataset analyzing?

Give me an overview of what this dataset is analyzing.

Generate key insights from this dataset

Lab 2: Create a New Julius AI Notebook with Your Own Dataset

Notebook link: https://julius.ai/s/notebooks/4bc1ff6e-af10-4b67-b389-1aa206bbef99?utm_source=share&utm_campaign=notebooks&utm_medium=referral

1. Find a Dataset

Dataset: Clothing.customers.csv

2. Analyze the Notebook

Summarize key trends:

- The average income is approximately \$72,618, with a wide range from 0 to over \$171,935, indicating some customers with very high incomes.
- The ages of customers range from 18 to 66 years, with a median age of 45, suggesting a fairly broad age distribution.
- The days since last purchase vary from 141 to 636 days, with a median of about 456 days, indicating that some customers haven't purchased in over a year.
- The annual spend averages around \$4,337, but there's significant variability, with some customers spending as little as \$243 and others up to \$7,572 annually.

Identify top spending customers in each segment

The top spending customers in each segment have been identified by selecting the customer with the highest annual spend within each cluster.

	cluster	income	age	days_since_purchase	annual_spend
38	0	102345	25	277	4696
768	1	58649	38	537	7406
668	2	40568	59	461	7572

- In segment 0, the highest spender has an income of approximately \$102,345, is 25 years old, and has spent about \$4,696 annually.
- In segment 1, the top spender earns around \$58,649, is 38 years old, and spends roughly \$7,406 annually.

- In segment 2, the leading spender has an income of about \$40,568, is 59 years old, and spends approximately \$7,572 annually.

Predict future spending based on past data

The code I ran trained a simple linear regression model to predict future annual spending based on three features: income, age, and days since the last purchase. I split the data into training and testing sets to evaluate how well the model generalizes to unseen data. After training, I predicted the spending on the test set and compared the predicted values with the actual values.

The table

 Export  Export  Expand 		
	Actual	Predicted
521	2859	2253.9687012894647
737	5775	5046.843159273478
740	5683	5288.49261756425
660	2613	2612.6937809145757
411	6143	4801.788348823518